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Unit 1 - Exploring the Core

PDF Documents

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This PDF document is included in the unit materials.

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UNIT 1 EXPLORING THE CORE

Topic Investigation



WHAT IS A FLOOD?

A flood is an overflow of water that submerges land that is usually dry. Floods can happen due to various reasons, including heavy rainfall, river overflow, dam failure, or rapid melting of snow. They can cause significant damage to property, disrupt communities, and harm the environment.

WHAT CAUSES FLOODS?

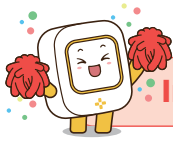
Floods can be caused by various factors, including heavy rainfall, river overflow, storm surges from hurricanes or typhoons, rapid melting of snow or ice, and dam failures. Human activities, such as deforestation and poor urban planning, can also contribute to the increased risk of flooding.



IMPACT OF FLOODS

Floods are natural disasters that can cause significant damage to the environment and communities. They can lead to loss of life, destruction of property, and disruption of ecosystems. Understanding the causes and effects of floods is crucial for developing effective prevention and mitigation strategies.





Importance of Flood Prevention

Preventing floods can help reduce their devastating impacts. Effective flood prevention strategies include building levees and dams, improving drainage systems, and using technology to detect and respond to rising water levels.

BUILDING LEVEES AND DAMS

- Levees are raised embankments constructed along rivers to prevent floodwaters from spilling over.
- Dams are barriers built across rivers to control water flow and store floodwater, releasing it gradually to prevent flooding downstream.



IMPROVING DRAINAGE SYSTEMS

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FLOODPLAIN MANAGEMENT

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USING TECHNOLOGY FOR EARLY WARNING

- Levees are raised embankments constructed along rivers to prevent floodwaters from spilling over.
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Questioning the Details

CASE STUDIES



RECENT FLOOD EVENTS

Q1 Can you think of any recent flood events in your area or around the world?
How did these floods impact the communities and environments affected?



SUCCESSFUL PREVENTION EFFORTS

Q2 Can you find examples of successful flood prevention efforts? What strategies and technologies were used to prevent or manage these floods?

QUESTIONING



Q3 How can technology help in flood prevention?



Q4 Why is sustainability important in preventing natural disasters like floods?



Inquiry & Reflection

Discussion question 1

Why is it important to understand natural disasters like floods and how to prevent them?

Discussion question 2

Reflect on the impact of floods on people, property, and the environment. How can knowledge and preparation make a difference?